

GJERGJ SHELDIJA

Lead Solution Architect · Clinical & MedTech Platforms · FHIR · Edge AI · Private Cloud · Regulated Environments

gjergj@sheldija.net · +49 176 710 82729 · sheldija.net · linkedin.com/in/gjergjsheldija · Leipzig, Germany
(open to European remote / DACH hybrid)

PROFILE

Solution Architect with 20+ years of hands-on experience designing and delivering complex, regulated clinical software ecosystems. Deep expertise in FHIR/HL7 interoperability, Edge AI infrastructure inside hospital networks, bare-metal Kubernetes on private datacenters, and medical device regulatory compliance (MDR / ISO 13485 / ISO 62304). Designed and delivered production Edge AI pipelines running clinical inference directly inside hospital networks without patient data leaving the secure zone. Engineered high-scale HL7 v2 → FHIR ingestion processing millions of messages per day. Built and currently redesigning a custom bare-metal Kubernetes private cloud targeting C5 certification — required because external cloud is not permissible under medical data regulations. Proven track record of translating architecture into measurable outcomes: 35% OpEx reduction, report generation from hours to under 2 minutes at Clinomic.

Open to Lead Architect, Principal Architect, and Solution Architect roles at MedTech, HealthTech, pharma, or biotech companies — remote-first or DACH hybrid, English-speaking.

TECHNICAL COMPETENCIES

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| ▶ FHIR R4 / SDC / Implementation Guides | ▶ Bare-Metal Kubernetes / Private Cloud | ▶ MDR / ISO 13485 / ISO 62304 |
| ▶ HL7 v2 & v3 / CDA | ▶ AWS (ECS / EKS / Lambda / S3 / RDS) | ▶ Compliance as Code |
| ▶ IHE Profiles | ▶ Docker / Container Orchestration | ▶ GDPR & Clinical Data Privacy |
| ▶ EHR / HIS Integration | ▶ Infrastructure as Code | ▶ C5 Certification (BSI) |
| ▶ OMOP Mapping | ▶ CI/CD & GitOps | ▶ Microservices & API-First |
| ▶ Mirth Connect (HL7 → FHIR) | ▶ Edge AI & On-Prem Inference | ▶ Keycloak / Identity Federation |

EXPERIENCE

Raylytic · Principal Architect | Engineering Lead *2017 – Present | Leipzig, DE (Remote)*
Progression: Senior Developer (2017) → Software Architect / AIQNET (2019) → Technical Lead (2022) → Principal Architect | Engineering Lead (2025)

Clinical data platform for orthopaedic and MedTech research. Regulated ISO 13485 / MDR / ISO 62304 environment.

Private Cloud & Kubernetes Infrastructure

- **Designed and delivered Raylytic's custom bare-metal Kubernetes platform** on private datacenter infrastructure — mandated because external cloud is not permissible under medical data regulations. Currently leading second-generation redesign targeting C5 certification (BSI), improved cost efficiency, faster disaster recovery and restores, horizontal scaling, and stable multi-environment deployment pipelines eliminating deployment risk across dev/staging/production.
- **Owns the full cloud strategy end-to-end:** architecture definition, infrastructure creation, application deployment, and migration — from initial build through current redesign.

Edge AI & Hospital Integration

- **Architected end-to-end Edge AI pipeline** enabling real-time clinical inference directly inside hospital networks — patient data never leaves the secure zone, eliminating the core data privacy constraint for AI-powered MedTech analytics.

- **Engineered high-scale HL7 v2 → FHIR real-time ingestion engine** (AIQNET platform) processing millions of messages per day, translating legacy hospital data streams into a fully standardised FHIR R4 ecosystem; implemented Mirth Connect unit testing pipelines for HL7 v2 → FHIR transformations.

Platform Architecture & FHIR Modernisation

- **Led multi-workstream platform transformation:** legacy monolith to FHIR-native microservices, EDC rebuild (MVP through production), OMOP mapping layer, patient data anonymisation, BI pipeline with real-time dashboards for medical researchers and hospital administrators.
- **Introduced FHIR SDC** for structured forms management across clinical workflows; integrated Keycloak as IdP for external identity federation; built FHIR IG management and validation pipelines.
- **Built end-to-end BI pipelines** delivering real-time dashboards for medical researchers and hospital administrators.

Regulatory Architecture

- **'Compliance as Code' CI/CD workflows:** automated MDR and ISO 13485 documentation and testing gates — ISO 62304-aligned software development lifecycle maintained without reducing delivery velocity.
- **Own architecture governance, decision records, and technology standards** across all engineering initiatives; translate engineering and regulatory risk into business language for board-level stakeholders.

FinOps

- **35% annual cloud OpEx reduction** via FinOps initiative on private infrastructure — documented against billing records; architecture optimisation converted directly into margin improvement.

Clinomic · System Architect | Strategic Infrastructure *Jan 2024 – Dec 2024 | Aachen, DE*
High-performance clinical data systems for intensive care environments. Returned to Raylytic at contract conclusion.

- **Reporting engine redesign:** reduced report generation from multiple hours to under 2 minutes — validated against internal system logs; adopted as company's long-term performance blueprint.
- **Led FHIR SDC transition,** eliminating legacy data silos and creating a unified cross-platform interoperability foundation across ICU clinical systems; updated FHIR Implementation Guide.
- **Designed and demoed a fully FHIR-native architecture** adopted as the company's forward strategy; created Mirth HL7 v2 → FHIR unit testing pipelines; added Keycloak as IdP; introduced FHIR SDC for form management.
- **Reduced systemic technical debt** through modular design patterns ensuring platform maintainability during rapid growth.

Sdai · Solution Architect *Jul 2023 – Dec 2023 | Germany (Remote)*
Infrastructure modernisation for a leading EdTech platform. Short engagement; returned to MedTech specialisation.

- **GitOps migration:** replaced fragile Bash-script deployments with declarative GitOps workflow — standardised releases, reduced failure rates.
- **Re-architected testing pipeline** for parallelisation, cutting regression feedback from hours to minutes.
- **Designed Unified Common Data Model** eliminating data silos across distributed services.

Trivago · Software Development Engineer *Jul 2018 – Jun 2020 | Leipzig, DE*
Full-stack engineering on the high-traffic Express Booking backend.

- **Led billing system migration from monolith to microservices on Kubernetes,** including full DevOps pipeline build-out and production monitoring — delivering deployment reliability and operational velocity at high-traffic commercial scale.
- **Optimised storage and distributed service integration** across the Express Booking platform.

UNICEF Albania · Lead IT Consultant *Dec 2017 – Apr 2018 | Tirana, Albania*
External standards authorship engagement. National Digitalization of Birth Records initiative, Ministry of Health.

- **Authored custom IHE Profiles for neonatal care**, establishing Albania's first standardised digital birth reporting protocol (HL7 v3 / CDA); architected HIS-to-State Health Registry secure integration layer.
- **Drafted Ministry of Health procurement roadmap** and technical Terms of Reference.

ONI (Hospital IT) · Head of Engineering (de facto CTO) 2007 – 2017 | Albania

Post-acquisition role following sale of Shkodra Software to ONI. Built entire technical organisation from scratch — 20 engineers + 2 POs across six major Albanian hospital deployments over 10 years.

- **Designed and delivered comprehensive HIS** across six hospitals including Mother Theresa, Trauma, and Shkodra — inpatient/outpatient, billing, pharmacy, surgery, laboratory; integrated with national eHealth platform via HL7 v3 and CDA.
- **Built custom CI/CD (Jenkins), Elastic search systems**, MySQL HA multi-master/multi-slave; rearchitected hospital network infrastructure; built and managed a custom private cloud solution.
- **Built and led a 22-person technical organisation from scratch** — engineers, network administrators, system managers, and product owners.
- **Led the Care2x open source Hospital Information System project for 4 years** — concurrent with ONI — delivering an international implementation in Tanzania and contributing to the global open source clinical community.

Shkodra Software · Co-Founder

2006 – 2007 | Shkodra, Albania

Co-founded Shkodra's first software house. Won Ministry of Health HIS contract within first year. Company acquired by ONI.

- **Founded and led technical delivery**; secured Ministry of Health contract within the first year — the commercial success that resulted in acquisition by ONI.

EDUCATION & CERTIFICATIONS

BSc Computer Science · Universiteti i Shkodrës Luigj Gurakuqi · 2006–2009

Health Information Exchange Certified Developer · **Health Informatics on FHIR** · **CHITP** · **BPM Living Application Developer** · **Healthcare Data Standards**

LANGUAGES

Albanian — Native · **English — Full Professional** · **Italian — Full Professional** · **German — Conversational** · **French — Basic**

PUBLICATIONS

Care2x Web Based Integrated Healthcare Information & Resource Planning · Gjergj Sheldija
The Implementation of Care2x in a Tertiary Hospital (60,000+ Patients)

Both publications document production HIS work led concurrently across Albanian hospitals (ONI) and the Care2x open source implementation in Tanzania.